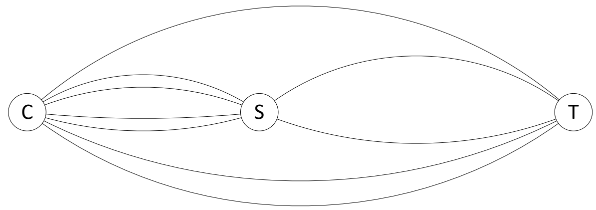
COMP 2121 **DISCRETE MATHEMATICS** | LAB 0

1. Three towns C, S, and T are interconnected as the figure below:



a) What is the number of different trips* from town C to town T?
*assume that a one-way trip includes any town at most once

$3 + 4(2) = 11$

b) What is the number of different round trips from town C to town T and back to town C?

$(11)(11) = 121$

c) How many trips from part b) include town S at least once? At most once?

at least once : one time, or two times

$4(2)(3) + 3(2)(4) + 4(2)(2)(4) = 112$

: all - none

$121 - (3)(3) = 112$

same

121		
S	S	S
none	one time	two times

at most once : none or once

$(3)(3) + 4(2)(3) + (3)(2)(4) = 57$

: all - two times

$121 - (4)(2)(2)(4) = 57$

same

2. In IPv4 address consists of 32 bits. They are canonically represented in dot decimal notation which consists of four decimal numbers, each ranging from 0 to 255, separated by dots, e.g., 142.232.55.77. Each address falls into one of the following classes:

- A class – used for the largest networks – begins with 0 which is then followed by a 7-bit network number, and then a 24-bit local address. Network numbers of all 0's or all 1's are not permitted.
- B class – used for the largest networks – begins with 10 which is then followed by a 14-bit network number, and then a 16-bit local address. Local addresses of all 0's or all 1's are not permitted.
- C class – used for the smallest sized networks – begins with 110 which is then followed by a 21-bit network number, and then an 8-bit local address. Local addresses of all 0's or all 1's are not permitted.

a) What class is the address 142.232.55.77?

1000 1110 class B

b) How many different addresses of each class are available in IPv4 standard?

all 0's ×
all 1's ×

class A: $0 \underbrace{7 \text{ bits}}_{N.N.} \underbrace{24 \text{ bits}}_{L.A.}$ $\boxed{(2^7-2)\cdot 2^{24}}$

all 0's ×
all 1's ×

class B: $10 \underbrace{14 \text{ bits}}_{N.N.} \underbrace{16 \text{ bits}}_{L.A.}$ $\boxed{2^{14}\cdot(2^{16}-2)}$

all 0's ×
all 1's ×

class C: $110 \underbrace{21 \text{ bits}}_{N.N.} \underbrace{8 \text{ bits}}_{L.A.}$ $\boxed{2^{21}\cdot(2^8-2)}$

3. What value does a variable *counter* have after the following code execution?

1
```
counter = 100
for i = 1 to 10 {
        counter = counter + 2
        for j = 1 to 20 {
                counter = counter + 3
                for k = 2 to 30 {
                        counter = counter + 4
                }
        }
}
```

$$\text{Counter} = 100 + \underbrace{(10)}_{i}(2) + \underbrace{(10)(20)}_{i,j}(3) + \underbrace{(10)(20)(29)}_{i,j,k}(4)$$

$$= 23{,}920$$

2
```
counter = 100
for i = 5 to 10 {
        counter = counter + 2
        counter = counter + 3
        for j = 15 to 20 {
                for k = 20 to 30 {
                        counter = counter + 4
                }
        }
}
```

$$\text{Counter} = 100 + \underbrace{6}_{i}(2) + \underbrace{6}_{i}(2) + \underbrace{6(6)(11)}_{i,j,k}(4)$$

$$= 1{,}714$$

3
```
counter = 100
a = 15
b = 20
for i = 1 to 10 {
        a = a + 2
        for j = 1 to 10 {
                b = b + 3
        }
counter = (i+3)*a + 5*i*b + 4
}
```

vs counter = counter + something

* only the last iteration

$$a = 15 + \underbrace{10}_{i}(2) = 35$$

$$b = 20 + \underbrace{10(10)}_{i,j}(2) = 320$$

$$\text{Counter} = (10+3)(35) + 5(10)(320) + 4 = 16{,}459$$

```
counter = 100
for i = 1 to 10 {
        counter = counter + 2
        for j = 1 to 10 {
                counter = counter + 3
                for k = j+1 to 10 {
                        counter = counter + 4
                }
        }
}
```
4

i=1

j	k	
1	2-10	9
2	3-10	8
3	4-10	7
4	5-10	6
5	6-10	5
6	7-10	4
7	8,9,10	3
8	9,10	2
9	10	1
10	—	

+ = 45

$$\text{Counter} = 100 + \underbrace{10}_{i}(2) + \underbrace{10(10)}_{i,j}(3) + \underbrace{10(\overbrace{45}^{j,k})}_{i,j,k}(4)$$

```
counter = 100
for i = 1 to 10 {
        counter = counter + 2
        for j = i+1 to 10 {
                counter = counter + 3
                for k = j+1 to 10 {
                        counter = counter + 4
                }
        }
}
```
5

$$\text{Counter} = 100 + \frac{10}{i}(2) + \frac{45}{i,j}(3) + \frac{120}{i,j,k}(4) = 735$$

i	j,k
1	8+7+6+5+4+3+2+1
2	7+6+5+4+3+2+1
3	6+5+4+3+2+1
4	5+4+3+2+1
5	4+3+2+1
6	3+2+1
7	2+1
8	1
9	—
10	—

+ = 120

i=1

j	k
2	3
2	4
⋮	
2	10

j	k
3	4
3	5
⋮	
3	10

j	k
4	5
4	6
⋮	
4	10

. . . .

j	k
8	9
8	10

j	k
9	10

8 + 7 + 6 + ... + 2 + 1

If all of the problems can't be done in the lab period, students can work on them on their own.